AQA 3.2.2 and 3.2.3 Tests for ions (required practical 4)

AQA A-level Chemistry · Revision questions · Sheet 6 of 6

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

Sheet 6 of 6

19 marks · about 25 minutes

Key information

- NH_4^+: warm with NaOH(aq), damp red litmus turns blue; $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$
- CO_3^{2-}: dilute acid gives effervescence; CO_2 turns limewater cloudy
- SO_4^{2-}: add HCl-acidified $BaCl_2$(aq) — white precipitate; the acid removes CO_3^{2-} first
- Halides with HNO_3-acidified $AgNO_3$(aq): Cl^- white, Br^- cream, I^- yellow; F^- no precipitate
- Silver halides with NH_3(aq): AgCl dissolves in dilute, AgBr only in concentrated, AgI in neither

Section A: Short-answer questions

01 Describe a test for sulfate ions in a solution. Give the observation for a positive result.

[2 marks]

02 A colourless solution gives a yellow precipitate with acidified silver nitrate. Give the formula of the precipitate and state the effect of adding concentrated ammonia.

[2 marks]

03 Which statement about barium sulfate is correct?

[1 mark]

A It is soluble in water at a temperature of 100 °C.
B It is used in medicine because it does not dissolve in body fluids.
C It is a pale yellow solid.
D It reacts with acidified barium chloride solution.

04 Give an ionic equation for the reaction of carbonate ions with hydrogen ions.

[1 mark]

05 A piece of strontium is added to an excess of water. Which observation is correct and complete?

[1 mark]

A Colourless solution
B Colourless solution and effervescence
C Dense white precipitate
D Dense white precipitate and effervescence

Section B: Longer questions

06 A student completes a series of test-tube reactions on two solutions, A and B. Each solution contains an unknown inorganic compound. The table shows the observations.

Test	Solution A	Solution B
Test 1 Add H_2SO_4(aq)	White precipitate formed	Effervescence
Test 2 Add HNO_3(aq) followed by $AgNO_3$(aq)	White precipitate formed	Effervescence

Test	Solution A	Solution B
Test 3 Add NaOH(aq) and warm gently. Hold a piece of damp red litmus paper over the end of the test tube.	No visible change	Red litmus turns blue

Show how the observations can be used to identify the inorganic compound in Solution A and the inorganic compound in Solution B. Give ionic equations for the reactions that occur.

[6 marks]

07 A student is provided with separate unlabelled samples of four different solutions for analysis. The four solutions are known to be ammonium nitrate, potassium sulfate, sodium carbonate and magnesium nitrate, but the student does not know which sample is which.

Outline a series of test-tube reactions that the student can use to identify each of these solutions. Include:

- the expected observations
- ionic equations for any reactions.

[6 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Sheet 6 of 6

Question	Marking guidance	Comments	Mark
01	Add (HCl-) acidified $BaCl_2$(aq) white precipitate	*Source: Practice question (written for this set)*	2
02	AgI no visible change / does not dissolve	*Source: Practice question (written for this set)*	2
03	**B**	*Source: 7404/1 Jun 2018 Q15*	1
04	$CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$	*Source: Practice question (written for this set)*	1
05	**B**	*Source: 7405/3 Jun 2025 Q14*	1
06	**Level 3 (5–6 marks)** All stages covered and generally correct and virtually complete (at least two from stages 1 and 2 and at least three from stage 3); coherent, logical progression. **Level 2 (3–4 marks)** All stages covered but incomplete / with inaccuracies, OR two stages covered and generally correct and virtually complete. **Level 1 (1–2 marks)** Two stages covered but incomplete / inaccurate, OR one stage covered and generally correct and virtually complete. **Level 0** Insufficient correct chemistry.	*Source: 7404/1 Jun 2025 Q06.1* **Stage 1 Solution A** 1a white ppt in Test 1 so Ba^{2+} (Sr^{2+}); 1b white ppt in Test 2 so Cl^-; 1c barium chloride / $BaCl_2$ ($SrCl_2$) **Stage 2 Solution B** 2a effervescence in Test 1 and/or 2 so CO_3^{2-} / HCO_3^-; 2b red litmus turns blue in Test 3 so NH_4^+; 2c ammonium carbonate / $(NH_4)_2CO_3$ (or NH_4HCO_3) **Stage 3 Equations** $Ba^{2+} + SO_4^{2-} \rightarrow BaSO_4$; $Ag^+ + Cl^- \rightarrow AgCl$; $CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$; $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$	6
07	**Level 3 (5–6 marks)** All stages covered and generally correct and virtually complete; well structured, no repetition, clear use of technical terms. **Level 2 (3–4 marks)** All stages covered but incomplete / with inaccuracies, OR two stages covered and generally correct and virtually complete. **Level 1 (1–2 marks)** Two stages covered but incomplete / inaccurate, OR one stage covered and generally correct and virtually complete. **Level 0** Insufficient correct chemistry.	*Source: 7404/1 Jun 2024 Q04* **Stage 1 Tests** 1a add NaOH; 1b add NaOH (and warm), hold damp red litmus at mouth of tube; 1c add nitric / hydrochloric acid; 1d add acidified $BaCl_2$ / $Ba(NO_3)_2$ **Stage 2 Observations** 2a Mg^{2+} white ppt with NaOH; 2b red litmus turns blue for NH_4^+; 2c CO_3^{2-} effervesces with acid / gas turns limewater cloudy; 2d SO_4^{2-} white ppt with $BaCl_2$; 2e final tube is the remaining solution **Stage 3 Equations** $Mg^{2+} + 2OH^- \rightarrow Mg(OH)_2$; $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$; $CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$; $SO_4^{2-} + Ba^{2+} \rightarrow BaSO_4$ To score 6 marks all 4 compounds must be identified	6